

$$\begin{array}{|c|c|} \hline 0 & x \\ \hline x^\top & 0 \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline u & u & 0 \\ \hline v & -v & v_0 \\ \hline \end{array} \begin{array}{|c|c|c|} \hline s & 0 & 0 \\ \hline 0 & -s & 0 \\ \hline 0 & 0 & 0 \\ \hline \end{array} \begin{array}{|c|c|} \hline u^\top & v^\top \\ \hline u^\top & -v^\top \\ \hline 0 & v_0^\top \\ \hline \end{array}$$